# AKLT model and Haldane phase

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February 15, 2019

# Outline


1. Spin projector Hamiltonian
2. Valence bond state: Dimer,V.B.Solid; V.B.Phase
3. Path integral of a single spin
4. Path integral of 1d AFM spin chain

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# Spin projector

Fuse two spin-1/2: $(1/2) \otimes (1/2) = (0) \oplus (1)$

- total-spin$(0)$: antisymmetric wavefunction $10 - 01$
- total-spin$(1)$: symmetric $11$, $10 + 01$, $00$

Casimir operator: $\hat{S}^2 = (\hat{\boldsymbol{S}}_1 + \hat{\boldsymbol{S}}_2)^2$

- For two spin-1/2: $\hat{S}^2 = 3/4 + 3/4 + 2\hat{\boldsymbol{S}}_1 \cdot \hat{\boldsymbol{S}}_2$
- In each total-spin subspace: $\hat{S}^2 = s(s+1)\boldsymbol{I} := x$

Def projector $P_s(\hat{\boldsymbol{S}}_1 + \hat{\boldsymbol{S}}_2)$: project out component whose total-spin $\neq s$

# Construct projector (two spin-1/2)

Construct projector out of $\hat{S}^2 := x$

- notice the fact that a projector: eig.val.=0 or 1.

For two spin-1/2:

| $s$ | $x$ | $P_1$ | $P_0$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 1 | 2 | 1 | 0 |

So we choose $P_1 = \frac{x}{2}$, $P_2 = \frac{x-2}{2}$, s.t. $P_1 + P_2 = 1$.
Plug in $x$ operator explicitly:

$$P_1 = 3/4 + \hat{\boldsymbol{S}}_1 \cdot \hat{\boldsymbol{S}}_2 \text{ AF-Heisenberg}$$
$$P_0 = 1/4 - \hat{\boldsymbol{S}}_1 \cdot \hat{\boldsymbol{S}}_2 \text{ F-Heisenberg}$$

# Projector Hamiltonian

One can construct spin lattice Hamiltonian $H = \sum_i a_i P_s(i, i+1, i+2...)$ where $a_i > 0$. If any state annihilate by all $P_s$, then it should be a ground state.

Example:

- spin-$1/2$ chain, n.n. projector: Heisenberg model
- spin-$1/2$ chain, n.n.n. projector: MG model(Majumdar-Ghosh)
- spin-1 chain, n.n. projector: AKLT model(Affleck-Kennedy-Lieb-Tasaki)
- ...(many different choices for spin, dimension, neighber, projector)

Some of them has known ground state.

# Example: Spin-1/2 chain $H_{MG} = \sum_i P_{\frac{3}{2}}(i, i+1, i+2)$

Since $(1/2) \otimes (1/2) \otimes (1/2) = (1/2) \oplus (1/2) \oplus (3/2)$, there are two possible projector $P_{\frac{3}{2}}$ and $P_{\frac{1}{2}}$

| $s$ | $x = s(s+1)$ | $P_{\frac{3}{2}}$ | $P_{\frac{1}{2}}$ |
|---|---|---|---|
| 1/2 | 3/4 | 0 | 1 |
| 3/2 | 15/4 | 1 | 0 |

So we choose $P_{\frac{3}{2}} = \frac{x-3/4}{3}$, $P_{\frac{1}{2}} = \frac{15/4-x}{3}$. Plug in $x = (\hat{\boldsymbol{S}}_1 + \hat{\boldsymbol{S}}_2 + \hat{\boldsymbol{S}}_3)^2$ we have: $H_{MG} = \sum_i \frac{1}{2} + \frac{2}{3}(\hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+1} + \hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+2} + \hat{\boldsymbol{S}}_{i+1} \cdot \hat{\boldsymbol{S}}_{i+2})$ Which is equivalent to: $H_{MG} = \sum_i \hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+1} + \frac{1}{2}\hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+2}$

# Example: Spin-1 chain $H_{AKLT} = \sum_i P_2(i, i+1)$

| $s$ | $x$ | $P_2$ | $P_1$ | $P_0$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 |
| 1 | 2 | 0 | 1 | 0 |
| 2 | 6 | 1 | 0 | 0 |

So we choose $P_2 = \frac{x(x-2)}{24} = 1/3 + \frac{1}{2}\hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+1} + \frac{1}{6}(\hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+1})^2$
Equivalently we can write:

$$H_{AKLT} = \sum_i \hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+1} + \frac{1}{3}(\hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+1})^2$$

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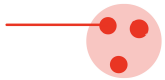
# Valence bond state

Valence bond (between two spin-1/2): $10 - 01$

- At most 1 bond per spin-1/2
- For N spin-1/2 system, each bond cost total spin by 1 unit

Valence bond show in figure as:

Symmetrize N spin-1/2 gives effectively one spin-N/2, show in figure as:

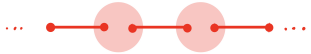


# V.B. state: Dimer and Valence bond solid

Ground state for $H_{MG}$: "Dimer"

- Periodic B.C., even sites: 2-fold.
- Open B.C., even sites: 5-fold; odd sites: 4-fold.
- Translational symmetry broken.
- No correlation beyond 2 site apart!

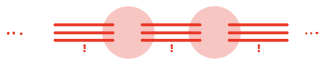
Ground state for $H_{AKLT}$: "Valence bond solid"



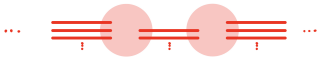
- Periodic B.C.: Unique.
- Open B.C.: 4-fold
- Translational symmetric
- Exponentially decay correlation

# Construct ground state for Spin s lattice

- If s $\in$ integer: n-VBSolid state $n < s$ is annihilated by projector $P_l(\hat{\boldsymbol{S}}_i + \hat{\boldsymbol{S}}_{i+1})$ as long as $l > 2s - n$



- If s $\in$ half-integer: Valance bonds are partially dimerized. Projector Hamiltonian with $P_l(\hat{\boldsymbol{S}}_i + \hat{\boldsymbol{S}}_{i+1})$ will be frustrated as long as $0 < l \leq \frac{2s+1}{2}$



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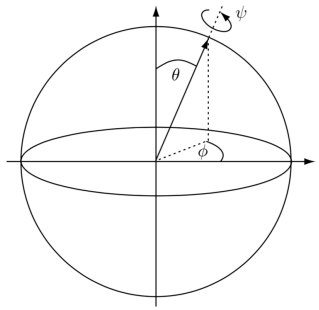
# Coherent spin state

A single spin can be characterized by 3 Euler angles:

$$
\begin{aligned}
|g(\theta,\phi,\psi)\rangle &= e^{-i\hat{s}^z\phi}e^{-i\hat{s}^y\theta}e^{-i\hat{s}^z\psi}|s^z=1\rangle \\
&= e^{-i\hat{s}^z\phi}e^{-i\hat{s}^y\theta}|s^z=1\rangle e^{-iS\psi}
\end{aligned}
$$




where $\phi \in [0, 2\pi), \theta \in [0, \pi)$, and $\psi$ is a phase factor due to gauge transformation. So we need only 2 parameters. Mathematically, $SU(2)/U(1) \simeq S^2$. Define a vector resembling a classical spin:

$$
\boldsymbol{n} \equiv \langle g(\theta,\phi,\psi)|\boldsymbol{s}|g(\theta,\phi,\psi)\rangle = S(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)
$$

# Path integral of a single spin

$H = \boldsymbol{B} \cdot \boldsymbol{s}$, $\boldsymbol{n}$ is moving on a sphere $S^2$ under magnetic field. To compute the partition function in imaginary time formalism, divide imaginary time interval $[0, \beta]$ into $N$ pieces of $\epsilon = \beta/N$ , and insert $N$ identities: $I = \int dg_i |g_i\rangle\langle g_i|$, and

$$\begin{aligned}\langle g_{i+1}|e^{-\epsilon H}|g_i\rangle &\approx \langle g_{i+1}|1 - \epsilon H|g_i\rangle = \langle g_{i+1}|g_i\rangle - 1 + 1 - \epsilon\langle g_{i+1}|H|g_i\rangle \\ &\approx \exp\left(\langle g_{i+1}|g_i\rangle - 1 - \epsilon\langle g_{i+1}|H|g_i\rangle\right) \\ &= \exp\left[-\epsilon\left(\frac{\langle g_{i+1}|g_i\rangle - \langle g_i|g_i\rangle}{\epsilon} + \langle g_{i+1}|H|g_i\rangle\right)\right]\end{aligned}$$

hence

$$Z = \int Dg \exp\left(-S[\theta, \phi]\right), \quad S[\theta, \phi] = \int_0^\beta d\tau[-\langle \partial_\tau g|g\rangle + \langle g|H|g\rangle]$$

# Wess-Zumino term

Apart from the contribution from Hamiltonian, the "kinetic energy" of a spin is described by Wess-Zumino action, or Berry phase action :

$$S_{WZ}[\theta,\phi] = \int_0^\beta d\tau \langle \partial_\tau g | g \rangle = iS \int_0^\beta d\tau (1-\cos\theta)\dot{\phi}$$

Since $\dot{\boldsymbol{n}} = \dot{\theta}\boldsymbol{e_\theta} + \dot{\phi}\sin\theta\boldsymbol{e_\phi}$, set $\boldsymbol{A} = (1-\cos\theta)/\sin\theta\boldsymbol{e_\phi}$ the action can be written as a line integral in parameter space:

$$S_{WZ}[\boldsymbol{n}] = iS \oint_\gamma \dot{\boldsymbol{n}} \cdot \boldsymbol{A} d\tau = iS \oint_\gamma \boldsymbol{A} \cdot d\boldsymbol{n}$$

We see that $\boldsymbol{A}$ can be seen as a vector potential in parameter space $S^2$, it's called Berry connection.

# Spin quantization

A vector potential ill-defined at south pole :

$$\boldsymbol{A}_N = (1 - \cos\theta)/\sin\theta \boldsymbol{e}_{\boldsymbol{\phi}} \quad \text{(singular point at } \theta = \pi)$$

A similar vector potential ill-defined at north pole $\theta \to \pi - \theta, \phi \to \pi + \theta$:

$$\boldsymbol{A}_S = -(1 + \cos\theta)/\sin\theta \boldsymbol{e}_{\boldsymbol{\phi}} \quad \text{(singular point at } \theta = 0)$$

The "magnetic field" is perpendicular to the sphere:

$$\boldsymbol{B} = \nabla \times \boldsymbol{A} = \boldsymbol{e}_r$$

# Spin quantization

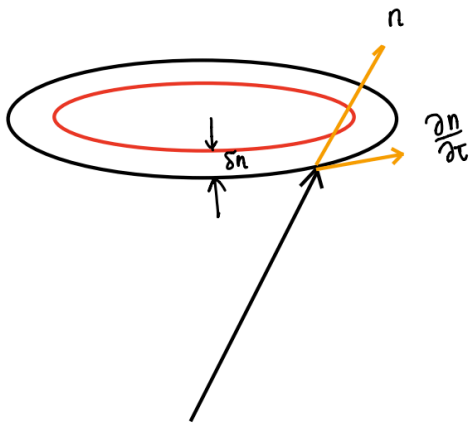
- We can apply Stoke's theorem:

$$\oint_{\partial\Sigma_N} \boldsymbol{A}_N \cdot d\boldsymbol{n} - \oint_{\partial\Sigma_S} \boldsymbol{A}_S \cdot d\boldsymbol{n} = \int_{\Sigma_N} \boldsymbol{B} \cdot d\boldsymbol{S} - \int_{\Sigma_S} \boldsymbol{B} \cdot d\boldsymbol{S} = 4\pi$$

- The action is the area enclosed by the closed path on the sphere
- Spin quantization is enforced by $S \in \mathbb{Z}/2 \Rightarrow e^{-S_{WZ}[\boldsymbol{n}]} = e^{-i4\pi S} = 1$

The deviation is just the area of the strip:

$$\delta S_{WZ}[\boldsymbol{n}] = \int_0^\beta d\tau \frac{\partial \boldsymbol{n}}{\partial \tau} \times \boldsymbol{n} \cdot \delta \boldsymbol{n}$$



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# Action of an anti-ferromagnetic chain

The Hamiltonian of AFM chain is

$$H = \sum_{\langle i,j \rangle} JS^2 \boldsymbol{n}_i \cdot \boldsymbol{n}_j$$

Write $\boldsymbol{n}_i$ into a Neel order parameter and a fluctuation term:

$$\boldsymbol{n}_i = (-1)^i \boldsymbol{\Omega}(\boldsymbol{R}_i) + a\boldsymbol{L}(\boldsymbol{R}_i)$$

They are perpendicular: $\boldsymbol{\Omega}(\boldsymbol{R}_i) \cdot \boldsymbol{L}(\boldsymbol{R}_i) = 0$

$$\begin{aligned} H &= JS^2 \sum_{\langle i,j \rangle} [(-1)^i \boldsymbol{\Omega}(\boldsymbol{R}_i) + a\boldsymbol{L}(\boldsymbol{R}_i)][(-1)^j \boldsymbol{\Omega}(\boldsymbol{R}_j) + a\boldsymbol{L}(\boldsymbol{R}_j)] \\ &\approx -JS^2 \sum_{i,\alpha} \boldsymbol{\Omega}(\boldsymbol{R}_i) \cdot \boldsymbol{\Omega}(\boldsymbol{R}_i + a\hat{\boldsymbol{e}}_\alpha) + JS^2 a^2 \sum_{i,\alpha} \boldsymbol{L}(\boldsymbol{R}_i)\boldsymbol{L}(\boldsymbol{R}_i + a\hat{\boldsymbol{e}}_\alpha) \\ &\approx -JS^2 \sum_{i,\alpha} \boldsymbol{\Omega}(\boldsymbol{R}_i) \cdot [\boldsymbol{\Omega}(\boldsymbol{R}_i) + a\frac{\partial \boldsymbol{\Omega}}{\partial R_\alpha} + \frac{a^2}{2}\frac{\partial^2 \boldsymbol{\Omega}}{\partial R_\alpha^2}] + 2JS^2 a^2 \sum_i \boldsymbol{L}^2(\boldsymbol{R}_i) \end{aligned}$$

# Action of an anti-ferromagnetic spin chain

$$= -\frac{JS^2a^2}{2}\sum_i \boldsymbol{\Omega}\cdot\nabla^2\boldsymbol{\Omega} + 2JS^2a^2\sum_i \boldsymbol{L}^2$$

$$= \frac{JS^2a^2}{2}\sum_i |\nabla\boldsymbol{\Omega}|^2 + 2JS^2a^2\sum_i \boldsymbol{L}^2$$

while the WZ action of each spin is:

$$S_{WZ}[\boldsymbol{n}_i] = S_{WZ}[(-1)^i\boldsymbol{\Omega}(\boldsymbol{R}_i) + a\boldsymbol{L}(\boldsymbol{R}_i)]$$

$$= S_{WZ}[(-1)^i\boldsymbol{\Omega}(\boldsymbol{R}_i)] + a\int_0^\beta d\tau \frac{\delta S}{\delta \boldsymbol{n}_i}\cdot\boldsymbol{L}(\boldsymbol{R}_i,\tau)$$

$$= (-1)^i S_{WZ}[\boldsymbol{\Omega}(\boldsymbol{R}_i)] + iaS\int_0^\beta d\tau \frac{\partial\boldsymbol{\Omega}(\boldsymbol{R}_i,\tau)}{\partial\tau}\times\boldsymbol{\Omega}(\boldsymbol{R}_i,\tau)\cdot\boldsymbol{L}(\boldsymbol{R}_i,\tau)$$

# Action of an anti-ferromagnetic spin chain

So the action of the chain is:

$$S[\boldsymbol{\Omega}, \boldsymbol{L}] = \sum_i S_{WZ}[\boldsymbol{n}_i] + \int_0^\beta d\tau H$$

$$= \sum_i (-1)^i S_{WZ}[\boldsymbol{\Omega}_i] + \int dx d\tau \left[ iS \frac{\partial \boldsymbol{\Omega}}{\partial \tau} \times \boldsymbol{\Omega} \cdot \boldsymbol{L} + \frac{JS^2 a}{2} (\frac{\partial \boldsymbol{\Omega}}{\partial x})^2 + 2JS^2 a \boldsymbol{L}^2 \right]$$

The WZ action has opposite sign at two neighboring sites, and

$$\sum_{i=1}^{N} (-1)^i S_{WZ}[\boldsymbol{\Omega}_i] = \sum_{k=1}^{N/2} S_{WZ}[\boldsymbol{\Omega}_{2k}] - S_{WZ}[\boldsymbol{\Omega}_{2k+1}]$$

$$= \frac{1}{2} \int dx \frac{\delta S_{WZ}}{\delta \boldsymbol{\Omega}} \cdot \frac{\partial \boldsymbol{\Omega}}{\partial x} = \frac{iS}{2} \int dx d\tau \frac{\partial \boldsymbol{\Omega}}{\partial \tau} \times \boldsymbol{\Omega} \cdot \frac{\partial \boldsymbol{\Omega}}{\partial x}$$

So we get in imaginary time, $Z = \int D\boldsymbol{\Omega} D\boldsymbol{L} \exp\left(-\int dx d\tau \mathcal{L}_E[\boldsymbol{\Omega}, \boldsymbol{L}]\right)$

# Lagrangian density

With Lagrangian density

$$\mathcal{L}_E[\mathbf{\Omega}, \boldsymbol{L}] = 2JS^2a\boldsymbol{L}^2 + iS\boldsymbol{L}\cdot\frac{\partial\mathbf{\Omega}}{\partial\tau}\times\mathbf{\Omega} + \frac{JS^2a}{2}(\frac{\partial\mathbf{\Omega}}{\partial x})^2 + \frac{iS}{2}\mathbf{\Omega}\cdot\frac{\partial\mathbf{\Omega}}{\partial\tau}\times\frac{\partial\mathbf{\Omega}}{\partial x}$$

Write in real time, $Z = \int D\mathbf{\Omega}D\boldsymbol{L}\exp\left(i\int dxdt\mathcal{L}[\mathbf{\Omega}, \boldsymbol{L}]\right)$, $x_0 = t, x_1 = x$

$$\mathcal{L}[\mathbf{\Omega}, \boldsymbol{L}] = -2JS^2a\boldsymbol{L}^2 + S\boldsymbol{L}\cdot\mathbf{\Omega}\times\partial_0\mathbf{\Omega} - \frac{JS^2a}{2}(\partial_1\mathbf{\Omega})^2 + \frac{S}{2}\mathbf{\Omega}\cdot\partial_0\mathbf{\Omega}\times\partial_1\mathbf{\Omega}$$

Integrate out the fluctuation field $\boldsymbol{L}$, we get $Z = \int D\mathbf{\Omega}\exp\left(i\int dxdt\mathcal{L}[\mathbf{\Omega}]\right)$

$$\boxed{\mathcal{L}[\mathbf{\Omega}] = \frac{1}{2g}(\frac{1}{v}(\partial_0\mathbf{\Omega})^2 - v(\partial_1\mathbf{\Omega})^2) + \frac{\theta}{4\pi}\mathbf{\Omega}\cdot\partial_0\mathbf{\Omega}\times\partial_1\mathbf{\Omega}}$$

where $g = 2/S$, $\theta = 2\pi S$, $v = 2aJS$ is spin wave velocity. This is called non-linear sigma model. The second term is often called "$\theta$ term".

# Effect of $\theta$ term

For an infinite long chain and infinite long period, the space time manifold is a sphere $S^2$, so $\mathbf{\Omega}(x,t)$ is a map from $S^2$ to $S^2$. In this case, the "$\theta$ term" simply means number of times a sphere wind another sphere:

$$Q = \frac{1}{4\pi}\int dxdt\, \mathbf{\Omega}\cdot\partial_0\mathbf{\Omega}\times\partial_1\mathbf{\Omega} \in \pi_2(S^2) = \mathbb{Z}$$

So it contributes a factor $e^{i\theta Q} = e^{i2\pi SQ}$ to each configuration.

- when $S$ is integer, $e^{i2\pi SQ} = 1 \Rightarrow$ fluctuations add up "constructively" $\Rightarrow$ no long range order, finite energy gap
- when $S$ is half-integer, $e^{i2\pi SQ} = (-1)^Q \Rightarrow$ fluctuations add up "destructively" $\Rightarrow$ long range order, gapless

"Haldane's conjecture"

# Summary

- The action of single spin contains a Wess-Zumino term as "kinetic energy", it's also a Berry phase term in Euler angle parameter space;
- At large $S$ limit, 1d AFM spin chain is described by non-linear sigma model. Apart from the contribution of spin wave term, there is another "$\theta$ term" which has topological nature;
- The order parameter in 1d AFM spin chain is a map from 1+1 d spactime sphere to the Bloch's sphere, this leads to interpretation of $\theta$ term integral as winding number;
- $\theta$ term produces "constructive/destructive interference" of fluctuation for integer/half-integer spin, resulting in gapped/gapless dispersion.

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